

Pankaj Thakur

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PROFESSIONAL SUMMARY

AI & Full-Stack Developer pursuing B.Tech in AI & Data Science at MITS Gwalior (2027). Specialises in **Generative AI**, **RAG**, and **multi-agent LLM systems** using **LangChain**, **LangGraph**, and **Groq**. Shipped 3 production-grade projects: a 4-module AI research platform (21 API routes, SSE streaming), a meeting intelligence platform (Whisper + LangGraph + RAG), and a full-stack parking SaaS (3-role RBAC, Razorpay, Socket.io). **Top 7.3% nationally in GATE CSE 2026**. Seeking **AI Engineer / SDE Internship**.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior

Sep 2023 – Aug 2027

B.Tech in Artificial Intelligence & Data Science

CGPA: 8.3 / 10

Relevant Coursework: DSA, DBMS, OOP, Operating Systems, Machine Learning, NLP, Computer Networks

TECHNICAL SKILLS

Languages	Python, C++ , JavaScript (ES6+), SQL, HTML5, CSS3
Frontend	React.js, Vite, Tailwind CSS, React Router DOM, Streamlit, Leaflet
Backend	Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, RBAC, Zod
Databases	MongoDB Atlas, SQLite, MySQL, ChromaDB (Vector Store)
AI / GenAI	LangChain, LangGraph, RAG, HuggingFace Embeddings, Groq, OpenAI Whisper, Prompt Engineering, Agentic AI, Multi-Agent Systems, NLP
Tools	Git, GitHub, Docker, Postman, Razorpay, Google OAuth, Socket.io, SSE, Tavily API, FFmpeg
Deployment	Vercel, Render, MongoDB Atlas, Streamlit Cloud

PROJECTS

Summly — AI Meeting Intelligence Platform | [GitHub](#)

May 2026

FastAPI | LangGraph | Groq | OpenAI Whisper | ChromaDB | FFmpeg | SQLite | React.js | Streamlit | JWT

- Built **end-to-end meeting intelligence pipeline**: **FFmpeg** extraction (16 kHz WAV) → **spectral noise cleaning** (5–10% accuracy gain) → **Whisper int8** transcription → **LangGraph + Groq** analysis extracting summaries, action items, decisions, and topic tags.
- Implemented **RAG-powered Q&A**: 500-char chunking → **sentence-transformer embeddings** → **ChromaDB top-3 semantic retrieval** → Groq-grounded answers over any meeting transcript.
- Engineered **JWT + bcrypt multi-user auth**, SQLite schema (meetings/intelligence/users), **yt-dlp YouTube ingestion**, follow-up email generation, and **20+ validated REST endpoints** with structured error handling.

SmartPark — Full-Stack Parking SaaS | [GitHub](#)

April 2026

React.js | Node.js | Express.js | MongoDB Atlas | Socket.io | Razorpay | JWT | Zod | Leaflet | Vercel | Render

- Architected production parking SaaS with **3-role RBAC** (Driver / Owner / Admin) and **concurrency-safe booking** via **MongoDB ACID transactions + pessimistic locking** (`findOneAndUpdate` inside session) preventing double-bookings under load.
- Engineered **23 RESTful endpoints** (auth, parking, booking, payment, analytics, admin) with **JWT**, **Helmet**, **rate limiting**, **XSS/Mongo sanitisation**; **real-time availability** via **Socket.io broadcast** on every booking confirmation.
- Built **weighted geo-ranking** ($\text{score} = \text{dist} \times 0.5 + \text{price} \times 0.3 + \text{avail} \times 0.2$) powering **\$geoNear Smart Recommendations**; integrated **Razorpay** → **HMAC-SHA256 webhook** and **Google OAuth**; owner earnings analytics.

ResearchOS — Multi-Modal AI Research Platform | [GitHub](#)

March 2026

FastAPI | LangGraph | LangChain | Groq | React.js | ChromaDB | SQLite | SSE | JWT | Tavily API

- Architected **4-module AI research platform** (Topic Research, PDF Chat, News Intelligence, AI Dashboard) with **21 RESTful API routes**, JWT-secured SPA, and **streaming-first SSE design**; deployed on Render & Vercel.
- Engineered **async generator-ThreadPoolExecutor SSE bridge** enabling **Groq (~500 tok/s)** without event-loop blocking; **Search→Read→Write** multi-agent pipeline delivers cited Markdown reports with quality scores.
- Implemented **7-stage RAG subsystem**: **HuggingFace all-MiniLM-L6-v2 (384-dim)** + **ChromaDB** per-UUID session isolation; custom **bind_tools()** loop bypassed hidden ReAct prompts causing Groq HTTP 400 errors.
- Integrated **Tavily News + Open-Meteo** into **LangChain multi-tool Dashboard Agent** with API key rotation & graceful fallback for zero-downtime LLM inference.

ACHIEVEMENTS & CERTIFICATIONS

- GATE CSE 2026** — Secured AIR 14,658 among 200,000+ candidates (**Top 7.3% nationally**), achieving a score of 426 with strong performance in Algorithms, DBMS, Operating Systems, and Computer Networks.
- LeetCode [Profile]** — Solved 120+ problems across arrays, trees, graphs, dynamic programming, hashing, and recursion with consistent Medium/Hard-level practice.
- DSA & Problem Solving | PW Skills [Credential]** — Completed coursework covering data structures, algorithms, problem-solving techniques, and time-space complexity analysis.
- Backend Development (Node.js & Express.js) | Udemy [Credential]** — Learned REST API design, server architecture, middleware patterns, authentication, authorization, and JWT-based security implementation.